

$$E = hf = hc / \lambda \rightarrow \lambda = hc / E = 6.63 \times 10^{-34} \times 3 \times 10^8 / 3.2 \times 10^{-19} = 6 \times 10^{-7} \text{ m} \rightarrow \text{Visible light}$$

30 Ans: B

$$KE = p^2 / 2m$$

$$\text{Momentum } p = (1.00 \times 1.6 \times 10^{-13} \times 2 \times 1.67 \times 10^{-27})^{1/2} = 2.311 \times 10^{-20} \text{ kg m s}^{-1}$$

$$\Delta p = 0.01 \times 2.311 \times 10^{-20}$$

$$\Delta p \Delta x = h$$

$$\Delta x = 2.87 \times 10^{-12} \text{ m}$$